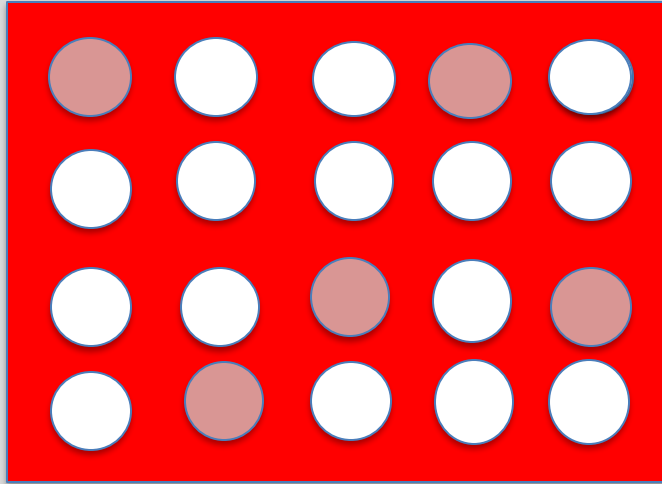


How does cardiac structure (histology) affect pathophysiology?

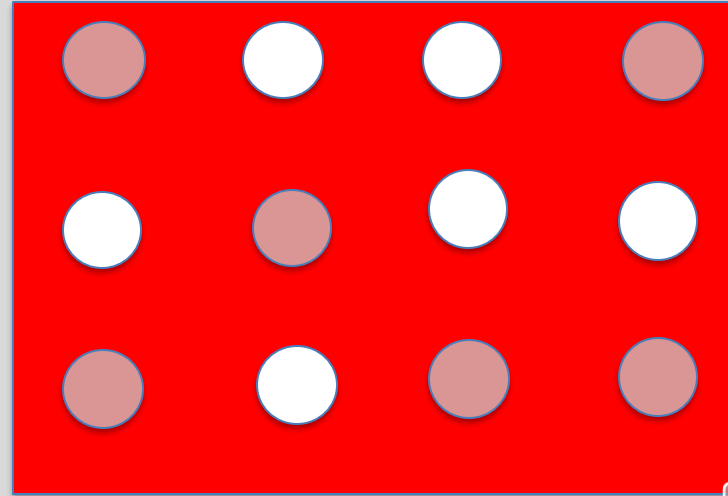
Cross sections of LV myocardium. Red areas packed with myocytes. Circles are capillaries. Myocytes and capillaries run in parallel. Pink = perfused, white = not perfused. LV normally has about 4,000 capillaries/mm² (left). Hypertension (right) > myocytes enlarge and push capillaries apart > ~2,500 capillaries/mm².

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine



Normal. ~1/4 capillaries perfused at rest. High capillary reserve.



Hypertension. ↓ capillaries, but more perfused (~1/2) to meet ↑ energy needs > ↓ capillary reserve > ↓ exercise capacity with ↑ chest pain on exertion.

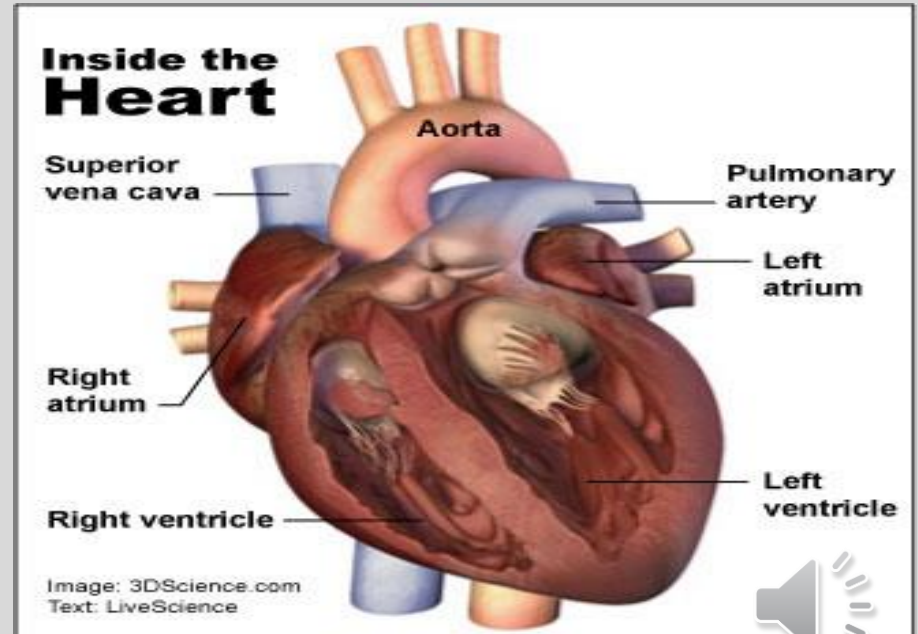


Normal heart data

- 250 - 300 gms (females)
- 300 - 350 gms (males, larger hearts due to larger body mass, the major regulator of heart mass)
- Left ventricle wall thickness: 1.3 – 1.5 cm
- Right ventricle wall thickness: 0.3 – 0.5 cm

NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine



Source: Chapter on Clinical Anatomy of the Heart from “Classic Teachings in Clinical Cardiology”, Ed. MA Chizner, 1996, Laennec Publishing Inc.

Interesting heart and vascular information

- Mammalian ventricles have ~25 M myocytes/gm (~7.5 B total in human heart)
- Myocyte proliferation ceases shortly after birth
- Continued heart growth is due to myocyte hypertrophy with no “proven” regeneration after this time. Non-myocytes continue to proliferate.
- There is considerable variation in myocyte size, but mean myocyte size in mammals is similar, with more myocytes in species with larger hearts.
- Males and females have about the same number of myocytes, but myocytes are slightly larger in males due to larger body weight.
- Approx. resting heart rates in beats/min (bpm): large whale (10), humans (70), dogs (90), rats (450), mice (600), hummingbird (1200).
- LV systolic pressure in giraffe: ~350mm Hg
- Giraffe's also have valves in their carotid arteries



Heart layers across the wall

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Endocardium

Endothelium, collagen, fibroblasts,
smooth muscle, nerves, purkinje
cells)

Myocardium

Myocytes, capillaries, blood vessels,
small nerves, lymphatics, minimal
CT mostly around vessels and
between myocytes

Thickness: LV > RV > atria

Epicardium

Fat cells, coronary vessels,
nerves, autonomic ganglia



Mesothelium

Pericardium (dense CT)

Pericardium

- Dense fibrous CT lined by mesothelium on inner surface.
- Acutely, pericardium does not expand (stretch).
- So, increased pericardial fluid can restrict heart function (cardiac tamponade).
- Examples: Pericarditis or chest trauma could fill pericardial cavity with blood).
- Note: First successful open heart surgery performed in Chicago in 1893. Dr Daniel Williams was African American and also established the first black owned hospital in the US. Knife wound to chest from bar fight.
- With chronic cardiac hypertrophy, pericardium can expand slowly, as you likely noted in gross anatomy lab.



Endocardium

NEW YORK INSTITUTE
OF TECHNOLOGY

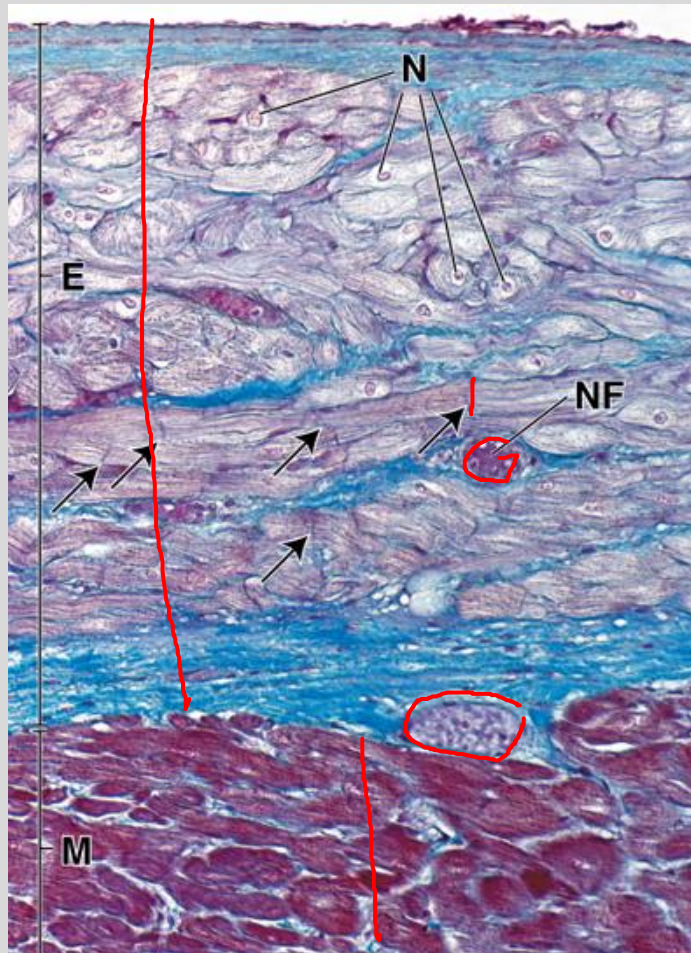
College of Osteopathic
Medicine

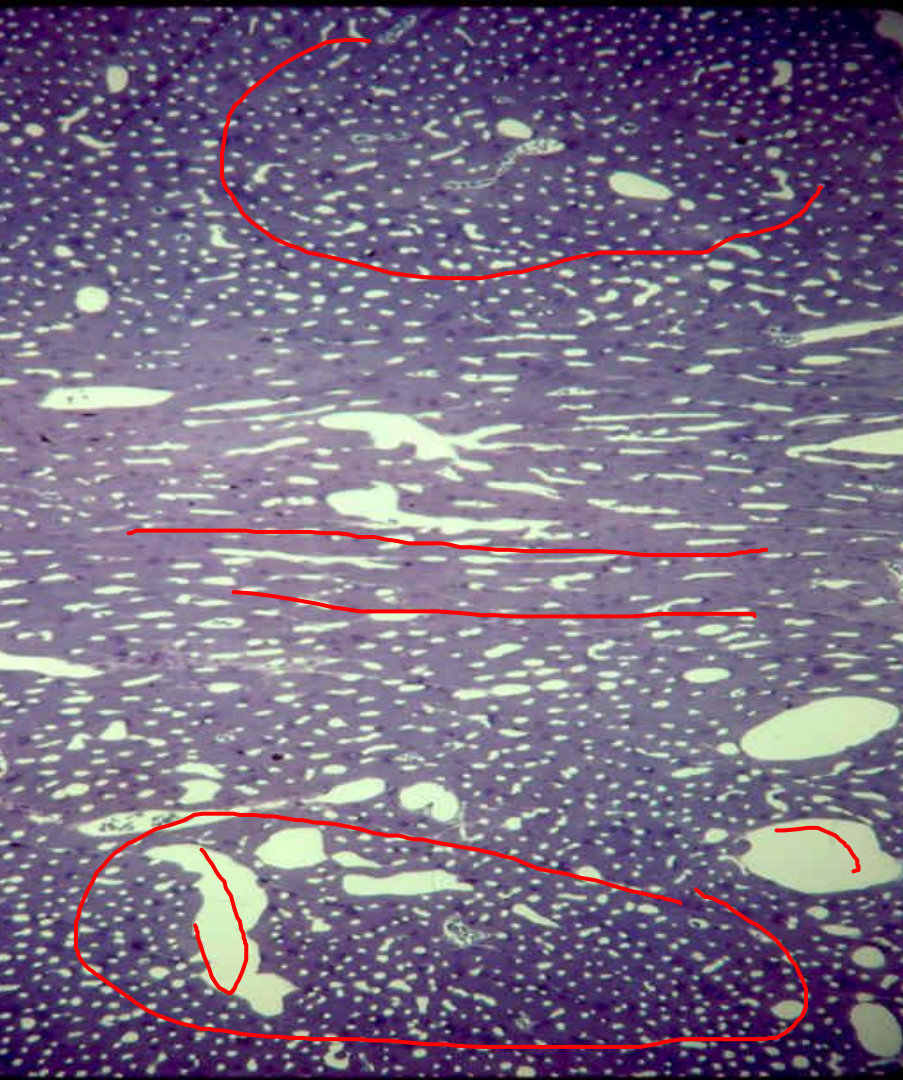
Ventricular cavity at top lined by endothelium. Blue staining is mostly collagen in subendothelial layer and throughout endocardium (E).

Upper 2/3 of image is mostly Purkinje cells. N, nuclei; NF, nerve fibers; arrows, intercalated discs.

Lower 1/3 is myocardium (M)

Source: Ross and Pawlina 13.10





NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic
Medicine

Myocytes change direction across wall somewhat like a Japanese fan (180 degrees from endo- to epimyocardium). Transmural section of rat RV perpendicular to base-apex axis.

Epicardial surface at bottom, endocardial surface at top. Myocytes (purple) and capillaries (small round white profiles) at top and bottom are cross-sectioned. Myocytes in middle are longitudinal or obliquely sectioned. Note: Rat heart is thin enough to show this in one field of view. Small white spots are patent capillaries. In adults, the ratio of capillaries/myocytes is approximately 1:1 (also true in humans).



Source: Gerdes

TEM Rat LV Myocardium (Cross Section)

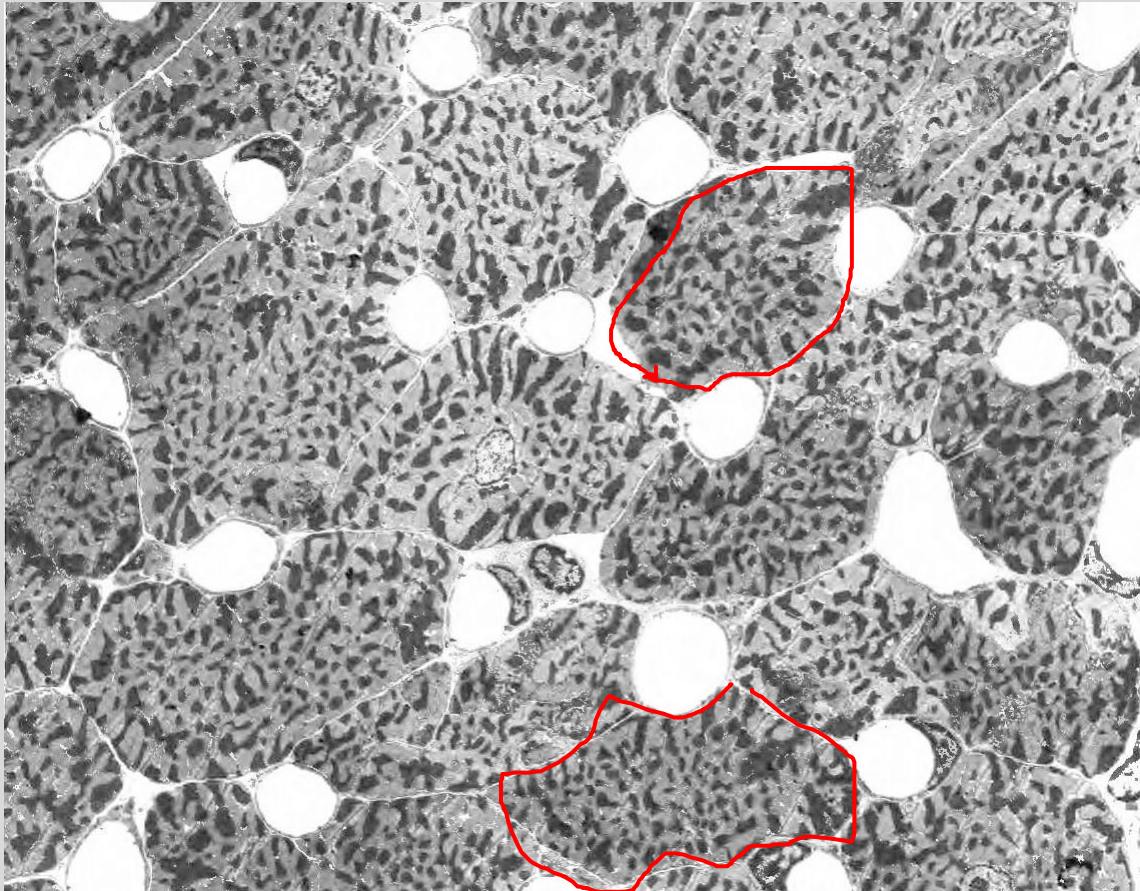
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Capillaries (clear circles) and myocytes (darker cells) cut in cross section. Darkest structures in myocytes, mitochondria, occupy $\sim 1/3$ of myocyte volume. Grey areas in myocytes are myofibrils. A central nucleus can be seen in some myocytes.



Source: AM Gerdes



Cardiac Myocytes

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Intercalated discs join cells, help synchronize myocyte contraction by allowing electrical coupling with relatively unrestricted ionic passage across the membranes of adjoining cells.

Cardiac muscle is specialized striated muscle, has more mitochondria (25-35% of cell) than skeletal muscle (3-5% of cell), and is very dependent on aerobic metabolism.

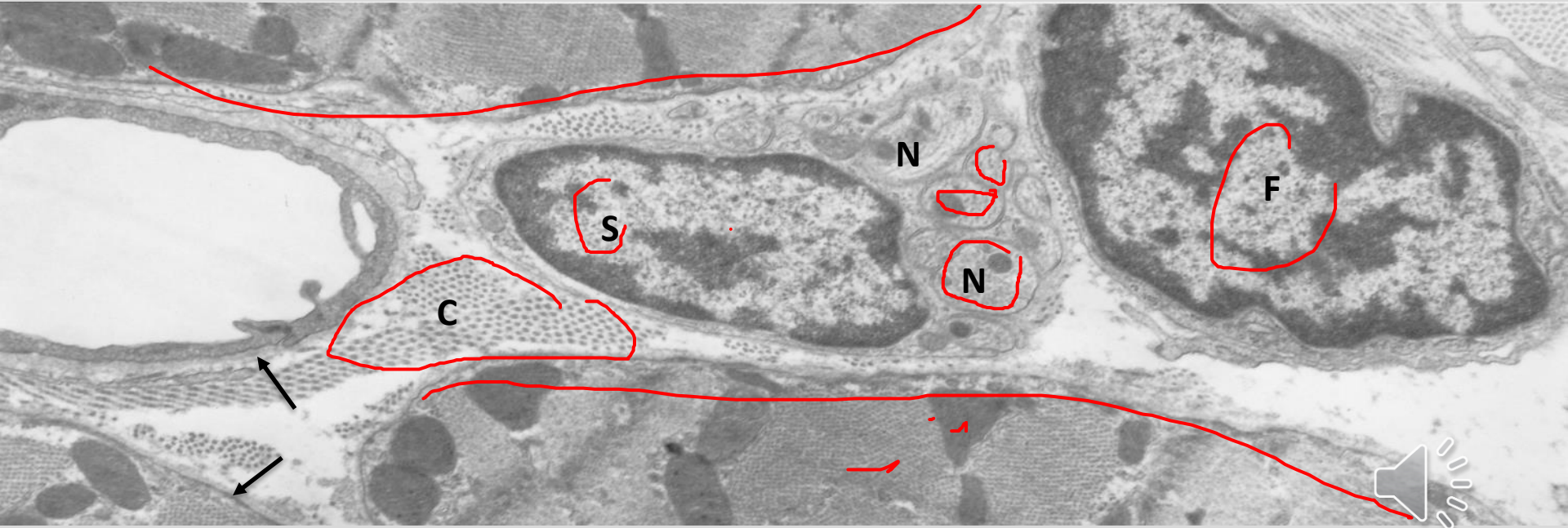


Papillary Muscles

- -Fingerlike projections from the inner surface of the heart connecting to chordae tendinae
- -Parallel myocytes and capillaries form the core
- -Covered by endocardium on outer surface
- -Have been used by animal physiologists to study cardiac muscle function *in vitro* (less complicated since fibers run in same direction)

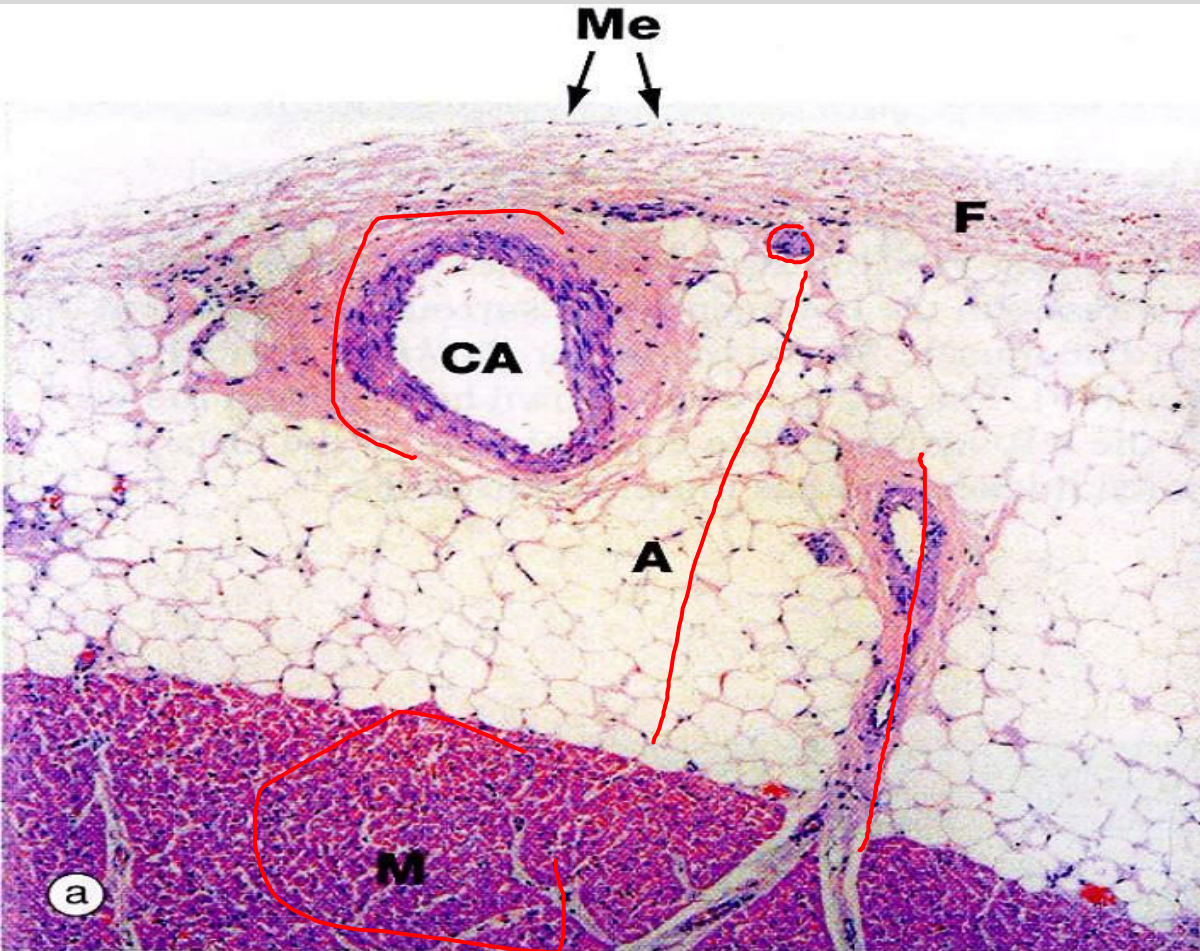


LV of rat showing nerves (TEM)



Schwann cell (S), nerve fibers (N) with mitochondria and neurofilaments, collagen fibers (C), fibroblast (F). External lamina (arrows) around myocytes and endothelial cells but not fibroblasts. Source: AM Gerdes

Epicardium



**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Epicardium:

- simple squamous epithelium or mesothelium (Me)
- elastin/collagen (F)
- Coronary artery (CA)
- adipose tissue (A)

Myocardium:

- Cardiac myocytes (M)



Source: Quizlet

Epicardium

- **Layer of mesothelial cells** on outer surface of heart with underlying connective tissue.
- **Blood vessels and nerves** (autonomic w postganglionic parasympathetic ganglia) which supply the heart are present in the epicardium
- Largely composed of **adipose tissue** which acts as cushion in the pericardial cavity.



Cardiac Skeleton

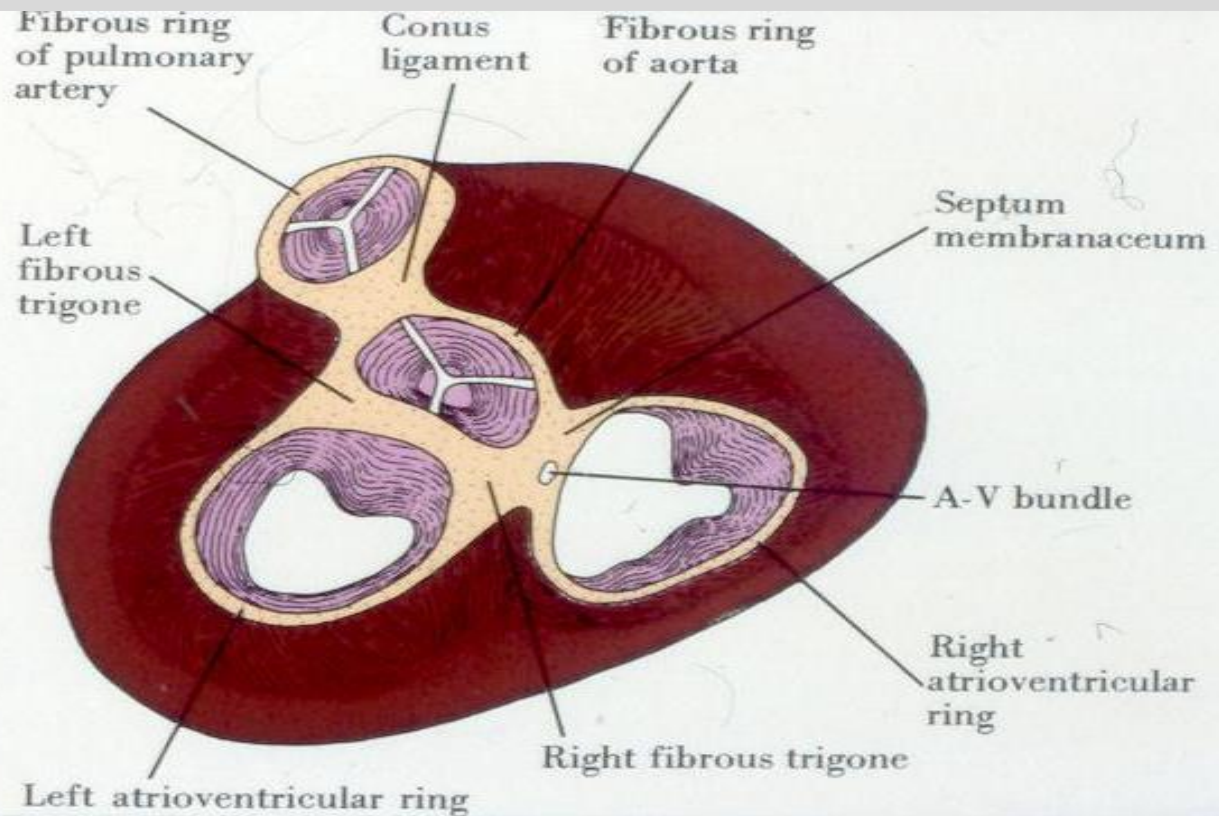
**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine

Dense fibrous CT
**surrounding 4
major valves,**
between valves
(trigones), and in
membranous
(upper)
**interventricular
septum.**



Source: Images.search.yahoo.com



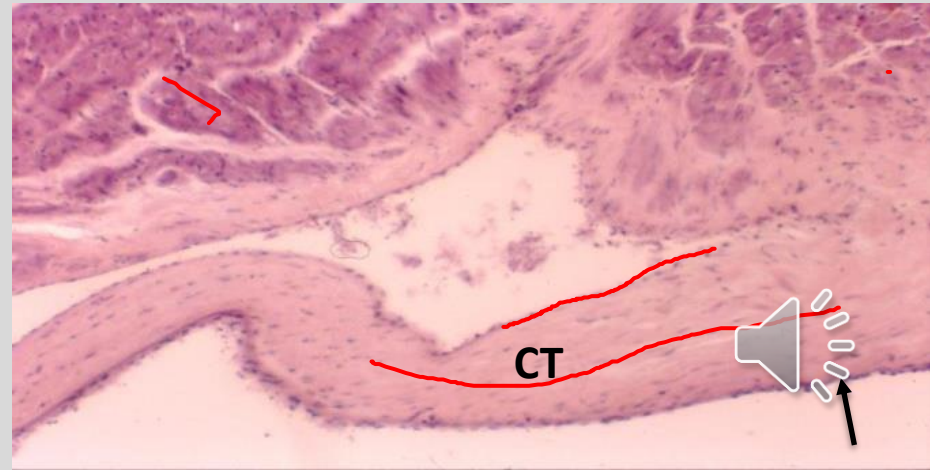
HEART VALVES and chordae (simplified)

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

- Valves: lined by simple squamous epithelium (endothelium, arrow) on both sides with dense connective tissue (CT) core extending from fibrous ring .
- Chordae tendinae: dense CT core covered by endothelium. (not shown)
- Avascular:
- get nutrition from blood
- circulating within chambers

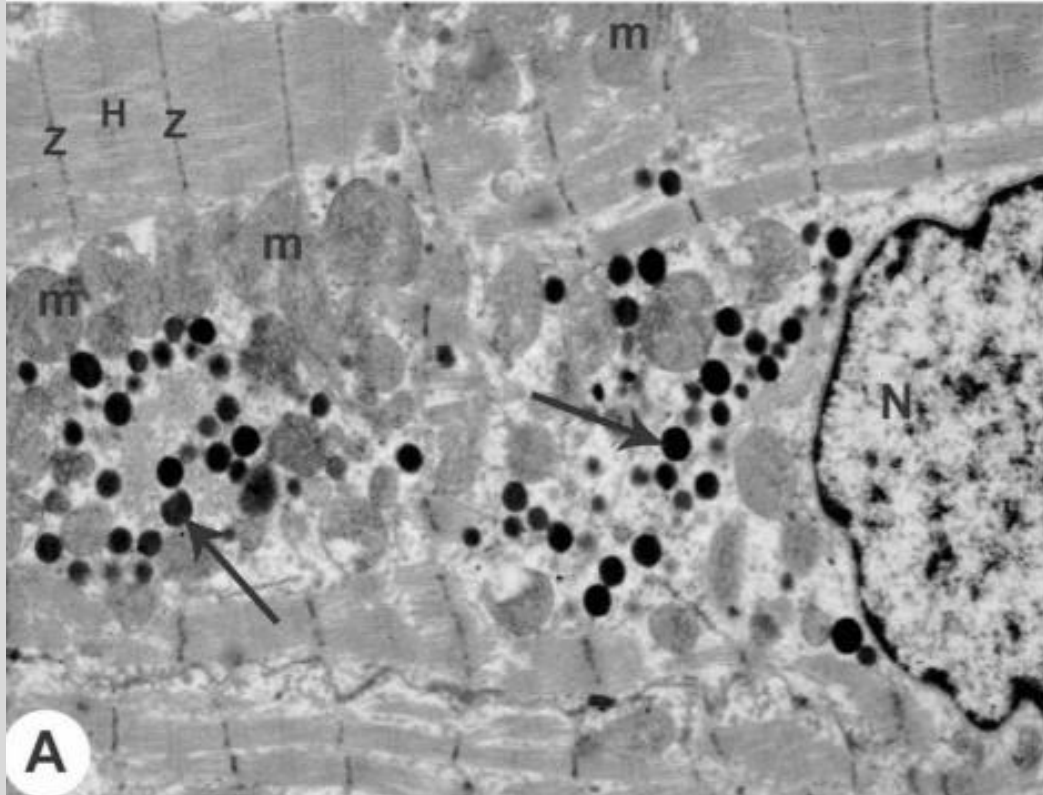
Source: Gerdes



Atrial Myocyte

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



ANP granules accumulate
at nuclear poles
Note: BNP is produced
primarily in ventricles,
especially in heart
disease, but not stored
(no granules in
ventricular myocytes)

Source: RA Eid, Int J Morphol
36:1031, 2018



Basics of cardiac pathology

NEW YORK INSTITUTE
OF TECHNOLOGY

College of Osteopathic
Medicine

Cardiac hypertrophy: In areas with viable myocardium, pressure overload leads to wall thickening from ↑ myocyte diameter. Chamber dilatation is due to ↑ myocyte lengthening from series sarcomere addition. So, physiological and pathological changes in ventricular anatomy are generally reflected by changes in myocyte shape. *AM Gerdes, 2002; J Cardiac Failure*

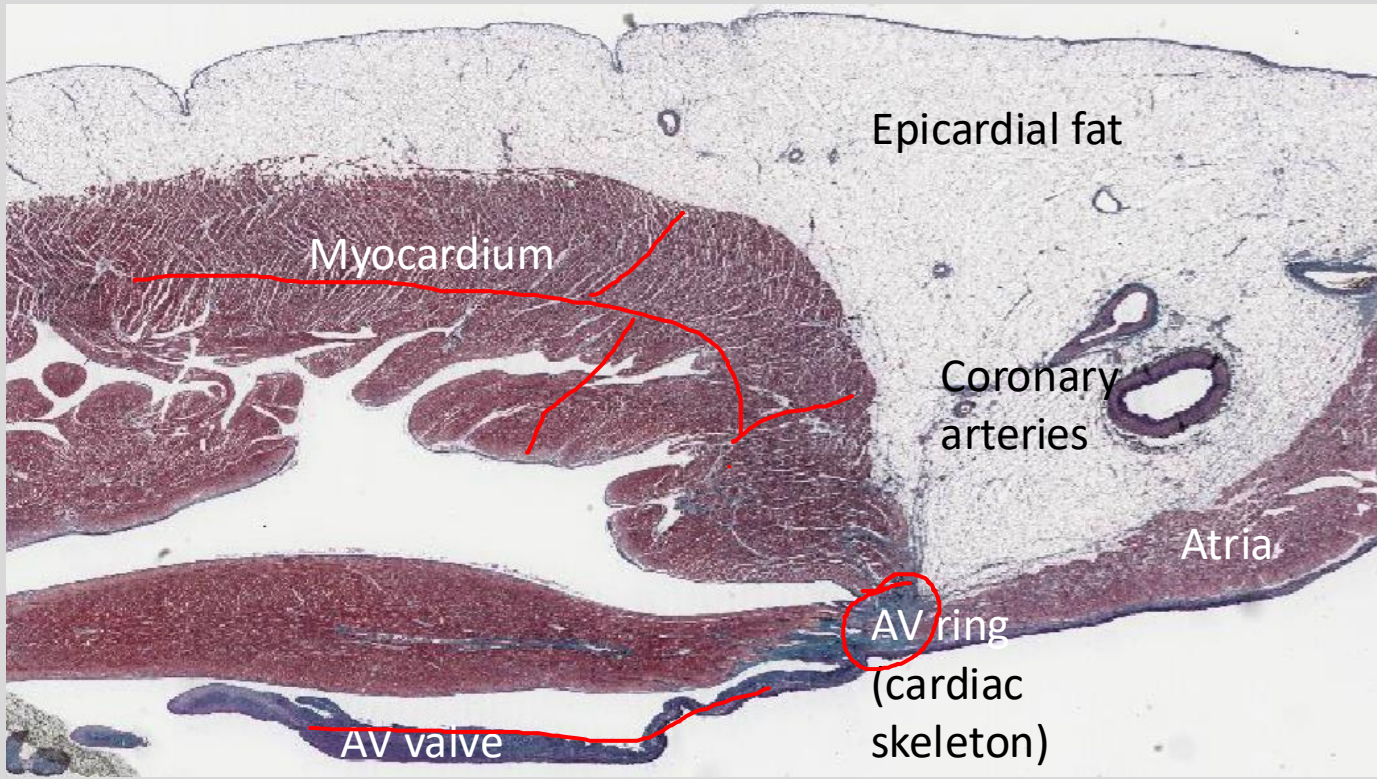
Myocardial fibrosis: Interstitial fibrosis between myocytes contributes to ventricular stiffness and impaired relaxation. Collagen has the tensile strength of steel! *Myocardial infarction* results in myocyte necrosis with replacement fibrosis. This looks like a large area of fibrosis where myocytes should be. Hypertension leads to perivascular fibrosis around arteries and contributes to vessel stiffness.



Heart: Low power view

**NEW YORK INSTITUTE
OF TECHNOLOGY**

College of Osteopathic
Medicine



Source: Univ Michigan
virtual histology

Summary Slide

- Students should: have a good mental image of the three layers of the heart and ventricular/atrial differences.
- Be able to describe how the morphologic composition of cardiac valves is quite functional.
- Be able to explain the basic morphology of the cardiac conduction system.
- Be able to identify various components of the cardiac skeleton and how their histological composition underlies function.
- Be able to describe the basic structure of the pericardium and how this composition underlies its function.

